Advanced Data Science with Gen AI

Innomatics Research Labs — Course Syllabus

NASSCOM FutureSkills Prime Certified Advanced Data Science with Gen AI. The official page describes a curriculum spanning Python, EDA/statistics, SQL, Power BI, machine learning, GenAI, deep learning, NLP, agentic AI, deployment and computer vision.

Source: https://www.innomatics.in/advanced-data-science-training-in-hyderabad/

Module 1: Python Programming

- Python fundamentals, Anaconda and Jupyter Notebook
- Identifiers, variables, data types, literals and keywords
- Operators and conditional statements
- while/for loops, comprehensions and problem solving
- Lists, tuples, sets, dictionaries and strings
- Functions, lambda, map/filter/reduce, modules and packages
- Object-oriented programming: classes, objects, inheritance, abstraction, polymorphism and encapsulation
- File handling and exception handling
- VS Code, Streamlit deployment and FastAPI basics
- AI-assisted coding with ChatGPT

Module 2: EDA with Applied Statistics

- EDA concepts and types of analysis
- Descriptive statistics, NumPy and Pandas
- Data manipulation, grouping, joins, missing values, strings and dates
- Matplotlib, Seaborn and Plotly visualization
- Probability and discrete/continuous distributions
- Sampling, Central Limit Theorem and confidence intervals
- Hypothesis testing, p-values, Type I/II errors, Chi-square and ANOVA
- Regex and ethical web scraping
- Data cleaning, preprocessing and EDA projects
- PandasAI and GenAI-powered data wrangling

Module 3: Database (MySQL)

- Database, DBMS, RDBMS, SQL vs MySQL/NoSQL
- DDL, DML, constraints and table creation
- Keys, relationships and ER diagrams
- SELECT, filtering, aggregation, GROUP BY, HAVING, ORDER BY and CASE
- Joins, UNION/UNION ALL and subqueries
- CTEs, views, stored procedures and window functions
- Transactions, ACID, triggers and date/time/string functions
- MySQL connectivity with Python
- Real-world SQL project

- GenAI-assisted SQL generation, optimization and debugging

Module 4: Reporting Tool (Power BI)

- Excel for data analysis, cleaning, lookups and pivot tables
- Power BI fundamentals and workflow
- Data import and visualization
- Power Query and data transformation
- Power Pivot, data modeling and relationships
- DAX: logical, text, mathematical, aggregation, filter and time-intelligence functions
- Dashboards, publishing, sharing and row-level security
- Drill-through, drill-down, interactions and Python visuals
- Power BI project and business reporting
- GenAI-assisted presentations and data storytelling

Module 5: Machine Learning Foundations

- AI vs ML vs Deep Learning; supervised vs unsupervised learning
- Regression vs classification and ML lifecycle
- Problem definition, features, target and objective setting
- Feature transformation, selection and importance
- Train/test split and data quality assessment
- Scikit-learn preprocessing, pipelines, training and prediction
- Regression metrics: MAE, MSE, RMSE, R^2 and adjusted R^2
- Classification metrics: confusion matrix, accuracy, precision, recall and F1
- Cross-validation and hyperparameter tuning with GridSearchCV/RandomizedSearchCV
- GenAI-enhanced ML case study

Module 6: GenAI Fundamentals for Data Analysts

- Generative AI vs traditional AI and analytics applications
- LLMs, context, prompts, responses, relevance and hallucinations
- Prompt engineering for data tasks
- GenAI for cleaning, EDA, code and SQL
- AI-assisted reporting, presentations and data storytelling
- Multimodal GenAI for text, images, tables, documents and charts
- Analytics workflow automation and n8n
- Responsible GenAI: privacy, security, bias and validation
- GenAI analytics case study and tool selection
- Synthetic data and AI image generation

Module 7: Data Analytics Projects & Capstone

- Retail Sales Analysis
- Student Performance Analysis
- PeopleOps Analytics
- HR Analytics: employee attrition and performance

- Pricing Strategy Analysis
- AI-Augmented Business Intelligence

Module 8: Applied Machine Learning

- Text preprocessing: tokenization, stop words, stemming and lemmatization
- Bag of Words and TF-IDF; spam/ham detection
- Image data and RGB channels
- Linear algebra foundations
- KNN for classification and regression
- Naive Bayes
- Decision Trees and impurity measures
- Linear and multiple linear regression, multicollinearity and VIF
- Polynomial regression and assumptions
- Gradient descent
- Logistic regression and classification
- Additional applied ML algorithms, ensembles and dimensionality reduction
- PCA and time-series forecasting
- AutoML and GenAI-assisted modeling

Module 9: Deep Learning and Artificial Neural Networks

- Biological neuron and artificial neuron
- Perceptron, weights, bias and activation
- MLP/FCNN/ANN and nomenclature
- Forward pass, backward pass and backpropagation
- TensorFlow/Keras model building
- Activation functions: linear, sigmoid, tanh, ReLU/LeakyReLU/PReLU and softmax
- Optimizers: gradient descent, mini-batch, momentum, adaptive methods, Adam and RMSProp
- Regularization: L1/L2, dropout, early stopping and batch normalization
- Optuna hyperparameter tuning
- End-to-end ANN project and Streamlit deployment

Module 10: Applied NLP with Transformers

- NLP preprocessing with NLTK/SpaCy
- BOW and TF-IDF
- Word2Vec and embeddings
- Text classification case study
- RNN, LSTM and GRU
- Sequence-to-sequence and attention
- Transformer architecture, positional encoding and multi-head attention
- GPT and BERT concepts
- Hugging Face API and embeddings
- Sentiment analysis, NER, POS tagging and QA
- Text generation, summarization and translation

- BLEU, ROUGE, METEOR and perplexity
- Hugging Face for image/audio tasks
- End-to-end NLP project

Module 11: Advanced Generative AI and Agentic AI

- LangChain, chat models and LCEL
- Prompt templates, chat prompts and message types
- Output parsers and basic chatbot
- Runnables and case studies
- Conversation memory, context optimization and summarization
- Monitoring and observability
- RAG: loaders, chunking, embeddings, vector databases and retrieval
- Hybrid RAG, reranking and metadata filtering
- RAG evaluation with RAGAS/LLM-as-a-Judge
- Tool calling, ReAct agents, custom tools and memory
- MCP architecture and integration
- LangGraph: state, graphs, sequential/parallel/conditional workflows
- Conversational agents and production-oriented agent systems

Module 12: AI Deployment, LLMOps and Production Systems

- FastAPI, Streamlit and Gradio for AI services
- API design, authentication and security
- Docker and deployment
- Azure AI Services and Azure AI Foundry
- Prompt management, model versioning, caching and cost optimization
- LLM application evaluation with LangSmith, RAGAS and DeepEval
- Hallucination detection and groundedness
- Logging, observability, tracing and performance optimization
- Scaling, enterprise deployment, CI/CD and multi-agent deployment

Module 13: Computer Vision

- Images, image storage and OpenCV
- Color spaces, bitwise operations, drawing, transformations and histograms
- Edge detection, blurring, video and webcam processing
- CNN fundamentals, convolution, padding, stride and pooling
- Image classification case studies
- Transfer learning with AlexNet, VGG16, Inception and ResNet
- Object detection: R-CNN family and YOLO
- Roboflow and YOLO case studies
- Image segmentation
- End-to-end computer vision project and Streamlit deployment

Module 14: Projects and Final Assessment

- Retail Sales Analysis
- Student Performance Management System
- PeopleOps Analytics
- Workforce Retention & Performance Analysis
- Retail Price Optimization
- ML Credit Risk Intelligence
- Automated Image Classification for Smart Logistics
- AI-Augmented Business Intelligence
- Enterprise AI Assistant

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